

Free Childcare and the Motherhood Penalty: Evidence from São Paulo

João Garcia Rafael Latham-Proença Marcela Mello Joana Silva

SECHI 2026, August 27

In Denmark, the child penalty explains nearly the entire long-run gender earnings gap (Kleven et al., 2019)

Can childcare close it? Evidence is mixed

Effects depend on the counterfactual: nulls come from high-coverage settings, where formal care crowds out informal (Havnes & Mogstad, 2011)

Latin America has the largest penalties, and scant evidence

We study a large expansion of free childcare in the city of São Paulo.

Large scale implementation

- Added on average 30,000 new seats per year over one decade
- Coverage went from 25% to 80%

Child penalty fairly typical of LatAm

Good data on labor and family structure

School district rules help with identification

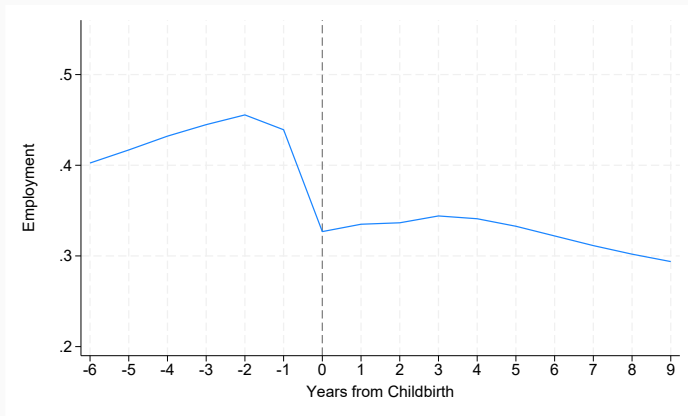
Question: How does free childcare availability affect the child penalty?

Method: We leverage the rollout of free childcare in São Paulo, in two complementary DID strategies

Results: An additional seat per child raises mothers' post-birth formal employment by 3.5 p.p. within districts, and by 6.8 p.p. between districts (about 20% of the post-birth mean)

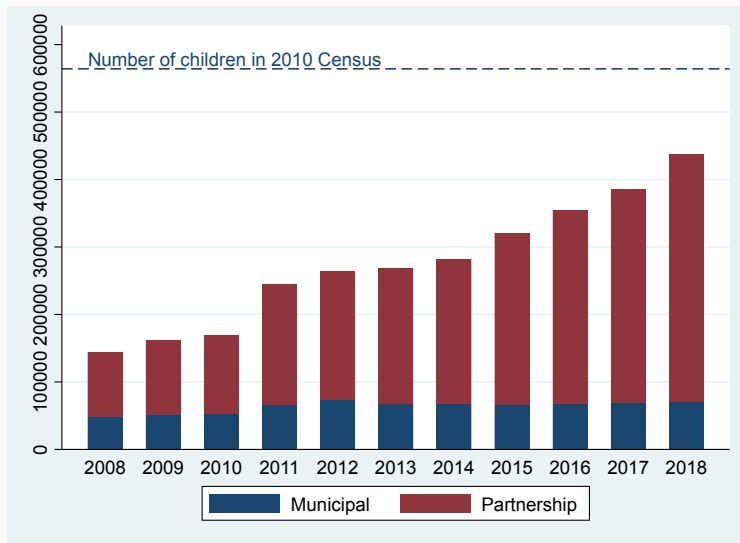
Setting

Setting: The Child Penalty in São Paulo



Formal employment falls 11 p.p. at the first birth, and does not recover

Setting: Enrollment



Setting: Public-private-partnership model

City government:

- Contracts with provider
- Provides suitable location
- Allocates families to centers

Private provider:

- Hires caretakers
- Handles day-to-day operation
- Helps find locations

High standard of quality

Flexibility enabled fast expansion



Childcare in Itaquerá, São Paulo

Allocation:

- Centralized online system to match child-facility

- First-come-first-served with a waitlist

- Preference for matching in the same district when available

- Priority for poorer families

Data

Sample: Family rosters and addresses from the **Single Registry**

Outcomes: Labor data from **RAIS**, includes all formal employment links

Treatment: Data on **childcare centers** from the City Government

Sample of

- Women
- Had their first child between 16 and 65 years old
- Living in the city of São Paulo
- In the Single Registry

Empirical Strategy

We use two complementary strategies, emphasizing different comparisons

Our **primary** strategy compares mothers and mothers-to-be **within** the same district, as availability changes

As a **complementary** strategy, we compare districts with and without a large expansion

Within-District Strategy

Within-District Strategy

We compare mothers and mothers-to-be in the **same district**, as childcare availability changes

Triple-difference structure, as in Kleven, Landais & Sørgaard (2020)

Why we take this as our primary design:

- Absorbs any local shock common to all women in the district

- Uses all the continuous variation

- Pre-birth coefficients are a built-in placebo

Within-District Strategy: Specification

Denote $Y_{d,t,\tau} = E[y_{i,t} | \text{district} = d, \text{time since childbirth} = \tau]$

$Availability_{d,t}$: total seats divided by children aged 0–3 in the district

$$Y_{d,t,\tau} = \gamma_{d,\tau} + \delta_{d,t} + \sum_{k \neq -2} \beta_k Availability_{d,t} \cdot 1_{\{\tau=k\}} + \varepsilon_{d,t,\tau}$$

Within-District Strategy: Fixing Ideas

Where does this come from? Suppose there is only one district, where we observe mothers and mothers-to-be

We can use a DID design, assuming mothers-to-be are not affected by childcare

- Two-by-two structure: mother/nonmother; before/after

With just 2 periods:

$$Y_{i,t} = \alpha + \beta \cdot \text{Availability}_t \cdot \text{Mother}_i + \gamma \text{Mother}_i + \delta \text{After}_t + \epsilon_{m,t}$$

We stack this over all districts, use time-from-childbirth in place of the motherhood indicator, and the full sample period

Within-District Strategy: Term by Term

Denote $Y_{d,t,\tau} = E[y_{i,t} | \text{district} = d, \text{time since childbirth} = \tau]$

$$Y_{d,t,\tau} = \gamma_{d,\tau} + \delta_{d,t} + \sum_{k \neq -2} \beta_k \text{Availability}_{d,t} \cdot 1_{\{\tau=k\}} + \varepsilon_{d,t,\tau}$$

Within-District Strategy: Term by Term

Denote $Y_{d,t,\tau} = E[y_{i,t} | \text{district} = d, \text{time since childbirth} = \tau]$

$$Y_{d,t,\tau} = \gamma_{d,\tau} + \delta_{d,t} + \sum_{k \neq -2} \beta_k \text{Availability}_{d,t} \cdot 1_{\{\tau=k\}} + \varepsilon_{d,t,\tau}$$

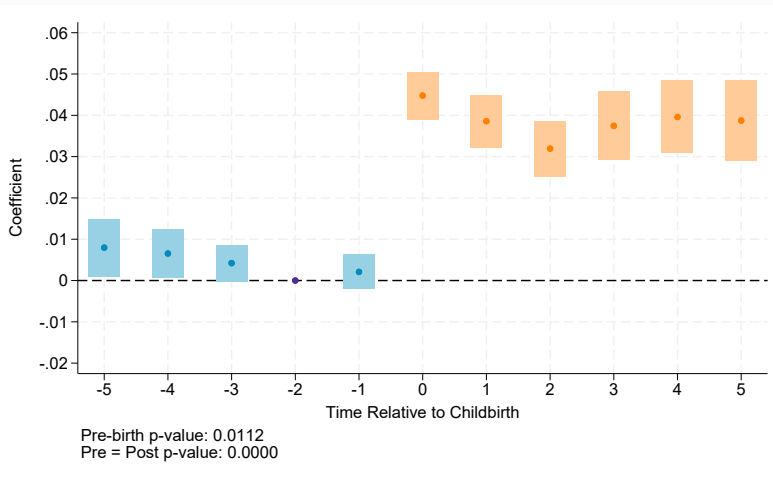
Within-District Strategy: Term by Term

Denote $Y_{d,t,\tau} = E[y_{i,t} | \text{district} = d, \text{time since childbirth} = \tau]$

$$Y_{d,t,\tau} = \gamma_{d,\tau} + \delta_{d,t} + \sum_{k \neq -2} \beta_k \text{Availability}_{d,t} \cdot \mathbf{1}_{\{\tau=k\}} + \varepsilon_{d,t,\tau}$$

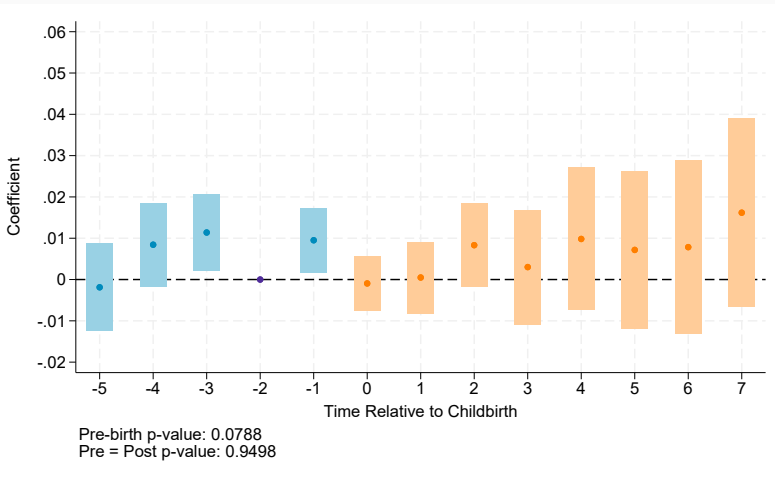
Within-District: Mothers' Employment

Mothers' Employment



Within-District: Fathers

Employment



Between-District Strategy

Between-District Strategy

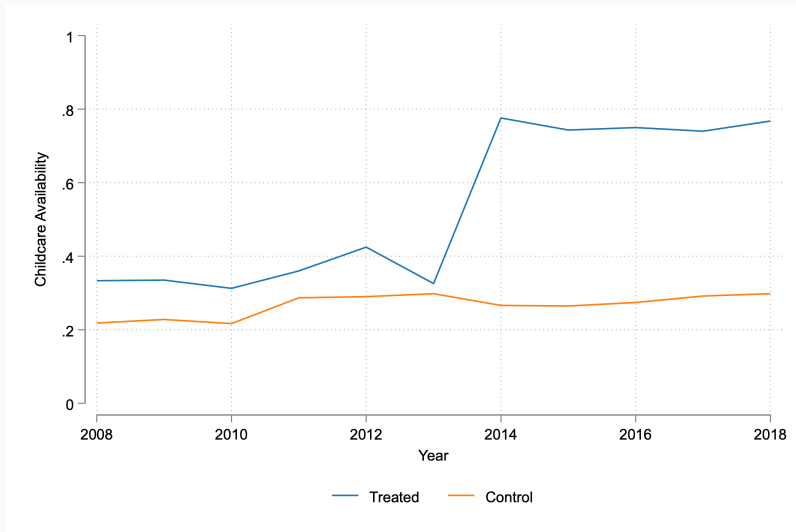
As a complementary strategy, we exploit the timing of large expansions across districts

Dynamic DID following Callaway and Sant'Anna (2021)

Identified off the **largest** expansions, so it speaks to a different margin

Costs: an “expansion” has to be defined, and parallel trends must hold **across** districts

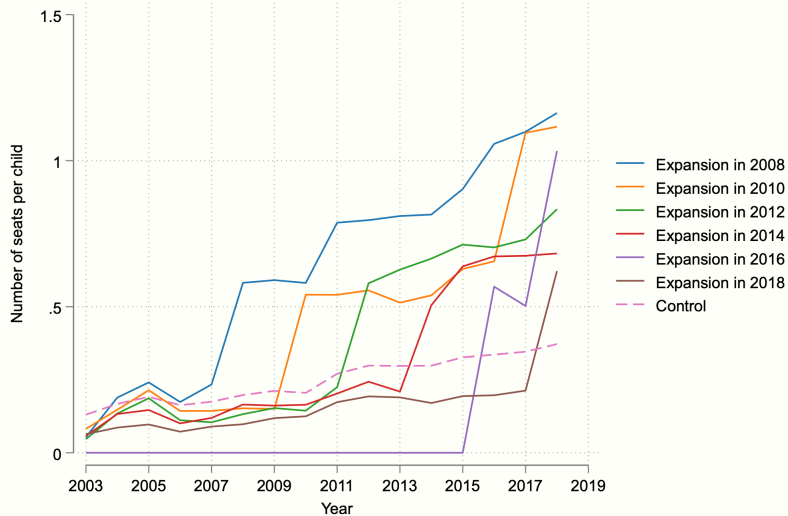
Between Districts



Between Districts: Defining an Expansion

- Calculate the largest annual increase in availability for each district over the period
- A district is considered **treated** if:
 - The first childcare facility opened during this period
 - Max increase is in the top 40% of distribution
- A district is in the **control** if $gmax_d$ is in the bottom 40%
- Districts with $gmax_d$ in the middle 20% are dropped from the comparison

Expansion of Childcare



Raw Before v After Comparison

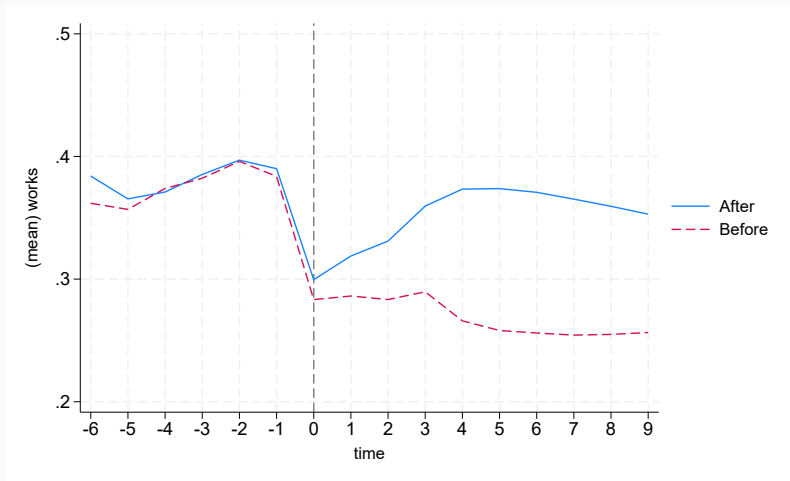
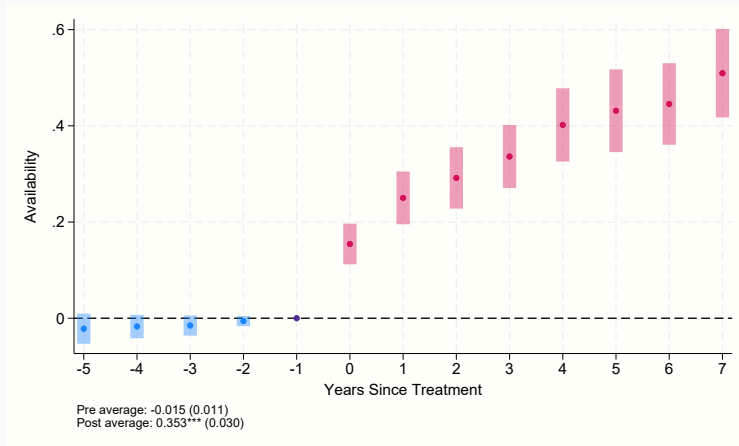


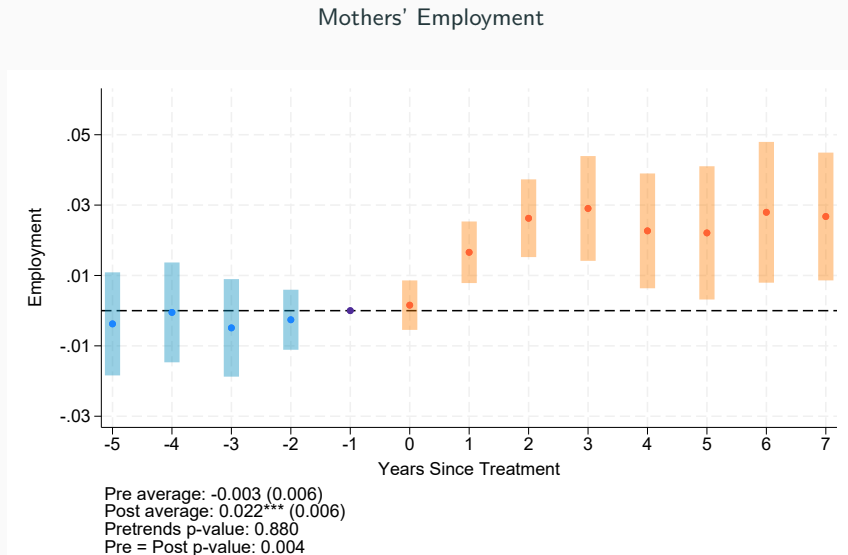
Figure 1: Child penalty for treated units comparing kids who were born before and after the expansion

Main Results: Childcare Seats per Child

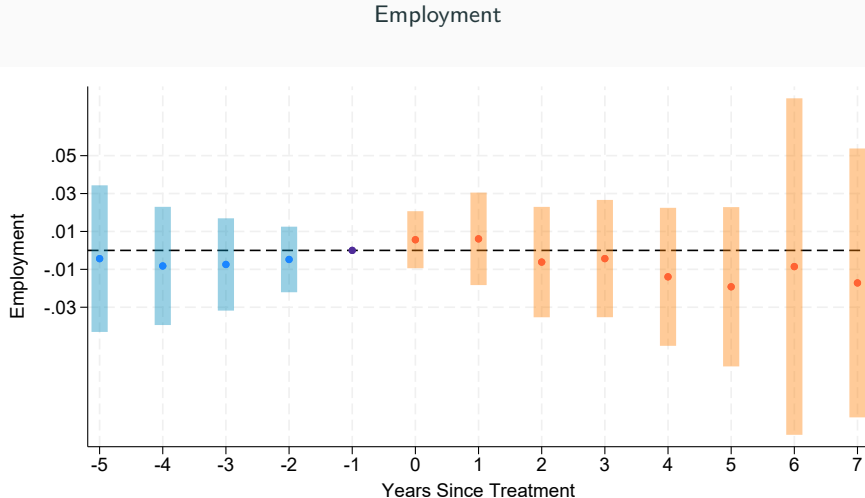
Childcare Seats per Child



Main Results: Mothers' Employment



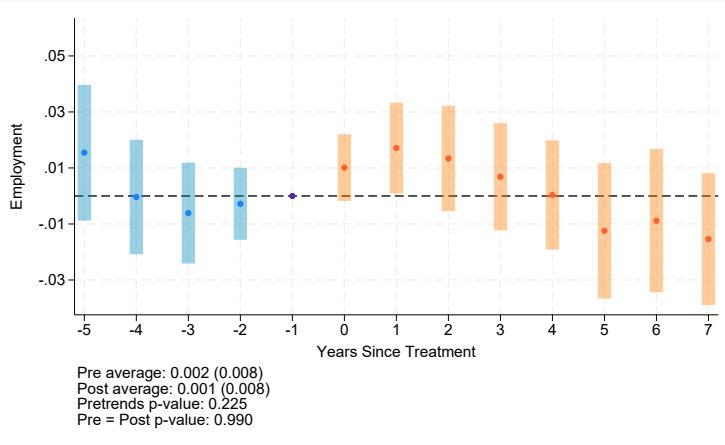
“Placebo”: Mothers-to-be Employment



Pre average: -0.006 (0.013)
Post average: -0.007 (0.015)
Pretrends p-value: 0.969

“Placebo”: Fathers’ Employment

Employment



Main Results: Between Districts

	Employment	Earnings	Wages
Mothers			
Pre	-0.005 (0.005)	-6.5 (7.8)	-4.5 (13.7)
Post	0.024*** (0.005)	28.8*** (8.4)	11.8 (15.4)
Post Rescaled	0.068	81.6	33.4
Future Mothers			
Pre	-0.006 (0.013)	-17.3 (22.9)	-48.9 (32.1)
Post	-0.007 (0.015)	-19.6 (20.9)	-49.7 (42.2)
Post Rescaled	-0.020	-55.5	-140.8
Fathers			
Pre	0.006 (0.018)	-10.9 (34.1)	-33.0 (71.1)
Post	0.008 (0.020)	-29.3 (29.6)	-137.3** (56.2)
Post Rescaled	0.023	-83.0	-389.0

"Post Rescaled" = effect $\div$ first stage of 0.353 seats per child. Monthly 2010 BRL.

Cheaper child rearing may also raise fertility, changing **who** the mothers are

An expansion raises births by 0.61 per thousand households per year

Against a sample mean of 10.8, this is a **5.7%** increase in fertility

Too small to drive the employment results: even if these “marginal” mothers were 17 p.p. more likely to work, the composition bias would be ≈ 1 p.p.

This paper studies the effects of free childcare on the motherhood penalty

We leverage a large scale expansion in São Paulo in complementary DID designs

We find one additional seat per child increases mothers' employment by 3.5 p.p. (within districts) to 6.8 p.p. (between districts), or roughly 10 to 20% of the post-birth employment mean.